MEDI-CINE
Vision Document
Group 04 | Online Cinema Management and Ticket Booking System
Version 1.2

Revision History

Date	Version	Description	Author
03/Oct/26	0.1	Initial project proposal and scope baseline	Group 04
11/Oct/26	1.0	Initial PA1 Vision Document aligned with the project plan	Group 04
13/Oct/26	1.1	PA1 review corrections: completed Product Perspective, aligned priorities, strengthened NFRs, and corrected document fields.	Group 04
22/Oct/26	1.2	PA2 Address the issues and making the documentation clearer and more detailed: - Removed the duplicated Problem Statement and Position Statement paragraphs - Expanded Alternatives and Competition with named competitors and a comparison table. - Added priority levels to Product Features.	Group 04

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Vision

1. Introduction

This Vision Document defines the product direction for MEDI-CINE, a web-based cinema management and online ticket-booking platform. It identifies the problem, business opportunity, target users, product value, scope, high-level features, priorities, quality goals, assumptions, and success criteria needed for PA1.

MEDI-CINE is intended for a mid-size cinema operation of approximately 15-20 screening rooms. It provides a customer-facing booking experience and an administration/staff dashboard. The product vision is deliberately high-level; detailed behavior will be refined through requirements, use cases, UX design, acceptance criteria, and later project increments.

1.1 References

MEDI-CINE Project Proposal (Group 04, version 0.1, 03 Oct 2026); MEDI-CINE Software Development Plan (Group 04, version 1.0, 13 Oct 2026); LN03 – Project Assignments (course material provided on Moodle, accessed 11 Oct 2026); PA1 Weekly Reports, Weeks 1 and 2 (Group 04, Oct 2026).

2. Positioning

2.1 Problem Statement

The problem of	inconvenient and fragmented cinema operations, including queues, schedule conflicts, duplicate seat sales, unclear payment status, and manual reporting.
affects	customers, cinema owners and managers, cinema staff, and administrators.
the impact of which is	slow or confusing ticket purchase, inconsistent operational data, booking errors, repeated ticket use, and avoidable manual work.
a successful solution would be	a single reliable workflow for movie discovery, scheduling, seat selection, booking, payment, QR admission, notifications, and reporting.

2.2 Product Position Statement

For	moviegoers and cinema teams who need convenient booking and dependable operational control.
Who	need to discover movies, choose showtimes and seats, complete payment, validate admission, and manage cinema data.
The (product name)	MEDI-CINE is a web-based cinema management and online ticket booking platform.
That	connects customer booking with staff QR validation and administrator control of movies, rooms, seats, schedules, pricing, and reports.
Unlike	manual counter workflows, spreadsheets, or disconnected tools that can produce queues, inconsistent data, and duplicate bookings.
Our product	provides one consistent, testable end-to-end workflow with server-side seat checks, confirmed-payment ticket issuance, and a foundation for controlled AI discovery.

3. Stakeholder and User Descriptions

3.1 Stakeholder Summary

Stakeholders include cinema owners or managers, cinema staff, customers, system administrators, the external payment provider, the project team, and evaluators.

Name	Description	Responsibilities
Cinema management / business owner	Owns the cinema operation and needs visibility into sales and operations.	Approves priorities and needs accurate schedules, prices, revenue, and manageable workflows.
Customer / moviegoer	Purchases tickets and attends screenings.	Needs easy discovery, clear seat availability, payment clarity, ticket access, and privacy.
Cinema staff	Validates admission at the cinema.	Needs fast QR lookup, clear validity results, and prevention of ticket reuse.
System administrator	Maintains accounts, roles, and operational data.	Needs access control, data consistency, and manageable administration.
Project team / evaluators	Builds and assesses the academic system.	Needs controlled scope, integration quality, test evidence, and demonstrable functionality.
Payment provider / AI service	External integrations used if supported by scope.	Needs secure request handling, authoritative backend data, and limited permissions.

3.2 User Summary

The main user classes are Customer, Cinema Staff, and Administrator. Project members and evaluators are additional stakeholders who need a demonstrable, tested system.

Name	Description	Responsibilities	Stakeholder
Customer	End user who discovers movies and books tickets.	Browse movies and showtimes, choose seats, pay, view booking history, and access e-tickets.	Cinema management / project team
Cinema Staff	Operational user who validates admission.	View showtimes, scan or enter QR codes, confirm ticket status, and record admission.	Cinema management
Administrator	Privileged user who maintains cinema data.	Manage users, movies, rooms, seats, showtimes, pricing, promotions, and reports.	Cinema management / project team
Project team / evaluator	Stakeholder and evaluator of the academic product.	Review scope, progress, test evidence, and final demonstration quality.	Course staff

3.3 User Environment

The user environment differs for each user class. The table below summarises the working context, task characteristics, platforms, and integrations that the design must support.

Aspect	Customer	Cinema Staff	Administrator
Typical location	Home, workplace, or on the move	Ticket counter and entrance lobby	Back office
Device	Phone, tablet, or desktop	Desktop browser with webcam/camera-based QR scan	Desktop browser
Browser /	Current Chrome, Edge,	Current Chrome, Edge, Firefox	Current Chrome, Edge,

platform	Firefox, and mobile browsers (responsive web)		Firefox
Main tasks	Browse movies, choose showtime and seats, pay, view e-ticket and history	Look up or scan QR, validate ticket, record admission	Manage movies, rooms, showtimes, pricing, promotions, users, reports
Task cycle time (target)	Booking completed in 5 minutes	QR validation within 2 seconds per ticket	3 minutes to create a showtime
People involved per task	1 customer (may book for a group)	1 staff member per check-in	1 administrator per change
Peak usage	Evening, weekends, new releases	Before each showtime	Off-peak hours
Environmental constraints	Variable network quality on mobile	Busy lobby, queues, possible low light	Controlled office
Account / access	Registered customer account	Staff role, restricted to validation functions	Admin role, authenticated and role-controlled
External systems	Payment sandbox, notification service	None directly	Payment sandbox (refund/status), reporting
Accessibility	Readable contrast, labels, non-colour-only seat cues	Clear valid/used/invalid results	Clear validation messages

All user classes work through HTTPS REST APIs against one backend, which remains the authoritative source for prices, seat availability, booking state, and ticket status.

3.4 Summary of Key Stakeholder or User Needs

Customers need clear now-showing and coming-soon movie discovery, search and filtering, understandable showtime and seat states, transparent prices, reliable booking and payment feedback, accessible e-tickets, booking history, and reminders before upcoming showtimes.

Cinema staff need fast QR lookup, clear paid/unused validation results, one-time admission control, and a simple way to prevent repeated use of the same ticket.

Administrators and cinema managers need consistent management of movies, rooms, seating layouts, showtimes, pricing, users, promotions, and revenue/occupancy reports, with role-based access to sensitive operations.

The project team and evaluators need a coherent end-to-end demonstration covering web development, database design, API integration, security, transaction handling, testing, deployment, documentation, and evidence that concurrent booking and payment failure cases are handled safely.

Priority: critical – booking correctness, seat availability, payment confirmation, live seat-state consistency, and one-time admission; high – movie discovery, seat selection, pricing, administration, reporting, notifications, reviews, and promotions; extension – richer reports, PWA support, and AI recommendations. The 14 core capabilities are the release baseline; optional extensions are considered only after the core booking path is stable.

The proposed solution replaces fragmented or manual workflows with a single system that validates schedules, protects seat availability, confirms payment before ticket issuance, and records auditable booking and admission states.

Need	Priority	Concerns	Current Solution	Proposed Solution	Acceptance Criterion
Correct seat availability and booking state	Critical	Concurrent booking and expired holds can cause duplicate	Manual counter tracking or none	Server-side availability checks, temporary holds, transaction	No duplicate seat booking under

		sales.		handling, and concurrency tests.	concurrent attempts.
Clear payment and ticket confirmation	Critical	Payment failure or duplicate callbacks can create unpaid or duplicate tickets.	Counter payment or disconnected provider flow	Explicit pending/success/failed/ expired states and QR ticket issuance only after confirmation.	No unpaid or duplicate QR ticket is issued.
Fast and reliable admission validation	High	Repeated ticket use slows entry and creates loss of control.	Manual ticket inspection	QR lookup, paid/unused validation, and one-time USED status.	A valid ticket can be used only once.
Convenient discovery and booking	High	Customers need clear movies, schedules, seats, prices, and booking history.	In-person counter or scattered information	Responsive web discovery, seat map, booking flow, e-ticket, and notifications.	A customer can complete the core booking journey.
Manageable cinema operations	High	Inconsistent operational data makes schedules, prices, and reports difficult to maintain.	Spreadsheets or separate tools	Role-controlled dashboard for movies, rooms, seats, schedules, pricing, promotions, and reports.	An administrator can maintain consistent operational data.
Controlled optional scope	Medium	AI and PWA extensions could consume time needed for the core booking workflow.	Not committed in PA1	Explore AI/PWA only after the 14 core capabilities are stable.	The core release remains complete without optional extensions.

3.5 Alternatives and Competition

MEDI-CINE serves a mid-size cinema operator of approximately 15-20 screening rooms. Its alternatives fall into four groups.

- Manual counter sales and spreadsheets: Familiar and free of software cost, but they cause queues at peak hours, duplicate seat sales, schedule conflicts, and manual reporting, and give no control against ticket reuse.

- Cinema-chain apps and websites (CGV Cinemas, Lotte Cinema, Galaxy Cinema, Beta Cinemas): Mature products with seat selection, online payment, vouchers, reward points, and ticket codes, backed by strong brand trust. Each is built for and owned by one chain, so an independent operator cannot configure its own rooms, seat layouts, pricing, promotions, or reports in them.

- E-wallet and aggregator channels (MoMo, ZaloPay): They offer a very large user base and convenient payment, and some partner chains can be booked through them. For an operator they mean dependence on a third party, possible fees, little control over how showtimes and prices are presented, and no operational tools such as room management, a dashboard, or QR admission control.

- Commercial cinema-management software (e.g., Vista Cinema, Veezi): The closest functional competitors, covering ticketing, scheduling, and reporting for exhibitors. They involve licensing and integration costs, are designed for broader markets, and cannot be inspected or modified, so they would not demonstrate the web-development, database, API, security, and testing work required by this course.

Criterion	Manual / spreadsheet	Chain apps (CGV, Lotte, Galaxy, Beta)	E-wallet channels (MoMo, ZaloPay)	Commercial software (Vista, Veezi)	MEDI-CINE
Online booking and seat selection	No	Yes	Partly (via partner chains)	Yes	Yes
Operator configures rooms, seats, pricing, promotions	Manually	No (chain-controlled)	No	Yes	Yes
Staff QR check-in with one-time use	No	Varies by chain	No	Yes	Yes
Duplicate-seat protection	No	Yes	Depends on partner	Yes	Yes (server-side checks and seat holds)
Revenue and occupancy reports for the operator	Manual	Not available to independent operators	No	Yes	Yes
Cost and dependency	Low cost, high labour	Tied to the chain	Fees and third-party dependence	Licence and integration cost	Own system, no licence fee

4. Product Overview

4.1 Product Perspective

MEDI-CINE is a client-server web application with a customer web interface and an administration/staff dashboard. Clients communicate with backend services over HTTPS REST APIs. Backend modules manage authentication, movies, cinema configuration, showtimes, seat holds, bookings, payments, tickets, QR validation, and statistics. A database stores operational records; a cache or realtime component may support temporary seat holds and live seat-state updates.

The project roadmap follows the Development Plan: Phase 1 Inception contains Sprint 1; Phase 2 Elaboration contains Sprints 2 and 3; Phase 3 Construction contains Sprints 4 and 5. Each sprint produces a demonstrable build. The project team consists of Trần Tấn Phát (Team Leader / Full-stack / DevOps), Huỳnh Vĩnh An (Full-stack / Data Engineer), Vân Hoàng Nhật (Secretary / AI Engineer / Frontend), Lê Hoàng Nam (Backend / Tester), and Nguyễn Hoàng Nam (Backend / Tester).

4.2 Assumptions and Dependencies

Assumptions and dependencies: the team has five members, access to a source repository and issue tracker, a database environment, representative cinema data, test browsers/devices, and a payment sandbox. The first release targets a responsive web application and desktop administration/staff dashboard, not a native mobile application. The final payment provider is not fixed; the payment adapter must isolate provider-specific behavior. AI/LLM availability is assumed only if the optional assistant is pursued. PWA support and AI assistance are optional extensions and may be dropped without affecting core booking. Changes in team capacity, schedule, provider availability, or TA feedback may require scope reduction.

5. Product Features

ID	Feature	Description	Priority
F-01	Authentication and authorization	Customer registration, login/logout, profile management, input validation, and role-based access for Customer, Staff, and Admin.	Critical
F-02	Movie catalogue management	Create, edit, deactivate, and maintain movie title, genre, duration, poster/trailer, age rating, and showing status.	High
F-03	Cinema room and seat management	Configure rooms, formats, seat layouts, seat types, and seat status.	High
F-04	Showtime scheduling	Create, update, activate, or cancel showtimes and validate room-time overlap.	High
F-05	Ticket pricing	Configure prices by seat type and day/time category.	High
F-06	Movie discovery	Browse now-showing and coming-soon movies, search/filter, view details, and find available showtimes.	High
F-07	Seat selection and temporary hold	Display a real-time seat map, re-check availability server-side, hold seats temporarily, and propagate seat-state changes.	Critical
F-08	Booking and payment	Create bookings, process payment through a sandbox gateway, and handle pending, success, failed, and expired states.	Critical
F-09	E-ticket and QR check-in	Issue a QR e-ticket only after confirmed payment, validate it for staff, and mark it USED after one admission.	Critical
F-10	Dashboard and reports	Show revenue, tickets sold, showtime performance, and seat occupancy.	High
F-11	Booking history	Let customers view current and previous bookings, status, showtime, seat, payment, and ticket details.	Medium
F-12	Reviews and ratings	Allow eligible customers to submit movie feedback.	Medium
F-13	Promotions and discount codes	Support promotional rules and discount-code validation.	Medium
F-14	Notifications and reminders	Send booking confirmations, payment updates, e-ticket notifications, and upcoming-showtime reminders.	Medium
F-15	AI movie/showtime assistant	Interpret controlled natural-language requests and recommend movies or showtimes using approved MEDI-CINE data. This	Optional / Planned AI

		feature is delivered only if time and stable APIs permit.	
F-16	PWA support	Allow the responsive web application to be installable on supported devices and provide limited offline-adjacent behavior or notifications if time permits. This is not part of the core PA1 commitment.	Optional

6. Non-Functional Requirements

Correctness and consistency: seat holds, bookings, payments, tickets, and check-in states must remain consistent under concurrent access. The system must re-check seat availability server-side, prevent duplicate seat sales, handle expired holds, and test duplicate payment callbacks. Acceptance criterion: concurrent attempts for the same seat must result in no more than one confirmed booking.

Security and privacy: enforce role-based access on backend APIs; validate inputs; use HTTPS in the deployed/demo environment where supported; protect credentials, payment secrets, and session data; restrict administrative operations by role; collect only information needed for accounts and bookings; and avoid sending unnecessary personal data to an optional AI service. Acceptance criterion: unauthorised roles are denied access to protected administrative operations in access-control tests.

Reliability and fault tolerance: handle payment failure, timeout, duplicate requests, expired holds, and failed callbacks without permanently locked seats or unpaid tickets. A QR ticket must be issued only after confirmed payment and must not be accepted more than once. Acceptance criterion: failed or timed-out payments do not produce a valid e-ticket, and a ticket marked USED is rejected on subsequent admission attempts.

Usability and accessibility: provide clear navigation, readable movie/showtime information, understandable seat-state indicators, price summaries, validation messages, payment/ticket status feedback, readable contrast, labels, and non-color-only seat cues. The customer flow should be usable on common phone and desktop screen sizes.

Compatibility, scalability, and performance: customer pages must be responsive across supported devices; admin/staff pages must support current Chrome, Edge, and Firefox. Provisional targets to validate during Elaboration: 95% of catalogue/showtime requests complete within 2 seconds, QR validation completes within 2 seconds, and the system handles at least 50 concurrent booking attempts without confirming the same seat for more than one booking. These targets are initial project acceptance goals and may be revised based on the agreed test environment and TA feedback.

Maintainability, observability, and documentation: separate customer UI, admin UI, API/business logic, persistence, payment integration, and optional AI/PWA integration into understandable modules; record meaningful state changes and errors without logging secrets; and provide setup, deployment, testing, role-based user, and demonstration documentation sufficient for course evaluation.